

PIERRE-LOUIS SIXDENIER

Embedded Systems Engineer • PhD Candidate

Nationality: French • pierre-louis.sixdenier@outlook.fr • +33 6 80 48 91 43 • github.com/sixdenip • linkedin.com/in/pierre-louis-sixdenier

EXPERIENCE

- [Job Title Placeholder • e.g. PhD Researcher / R&D Engineer]** **November 2020 – May 2026**
[Company / Research Institution / University Lab Placeholder] France
• *[Placeholder for research activities, engineering deliverables, or doctoral thesis work between Nov 2020 – May 2026].*
- FPGA R&D Engineer Intern** **May 2020 – Nov 2020**
Safran Electronics & Defense France
• Conception of a debugging streaming IP for an embedded low-power FPGA platform.
- Research Intern** **December 2018**
ETIS Laboratory (Equipes Traitement de l'Information et Systèmes) Cergy, France
• Conception and implementation of a failure detection system for robots.
- Research Intern** **June 2017**
XLIM-SIC Laboratory Poitiers, France
• Conception of a proof-of-concept (PoC) on an L-System generator.

EDUCATION

- Doctor of Philosophy (Ph.D.) in Computer Science** **2020 – 2026 (Exp.)**
[Doctoral School / Research Lab Placeholder] France
• Research in Embedded Systems, Hardware-Software Co-Design & Energy-Harvesting Edge Computing.
- Master in Computer Science** **2018 – 2020**
CY Cergy Paris University Cergy, France
• **Top-ranking of the research program in Smart Electronic Systems.** Advanced curriculum in FPGA, VHDL, and embedded systems.
- Exchange Program • Computer Science** **2017 – 2018**
Oregon State University Corvallis, OR, USA
• Followed CS classes and gave French lessons to undergraduate students.
- Licence in Computer Science** **2015 – 2018**
University of Poitiers Poitiers, France
• Graduated with high honors.

PUBLICATIONS (PEER-REVIEWED)

- [1] **Early-Exit Neural Architecture Search for Energy-Harvesting Edge Computing.**
P.-L. Sixdenier, M. Deutel, J. Teich. *IEEE 18th Int. Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoc 2025)*. **Best Paper Award.**
- [2] **Early-Exit Forecasting of Deep Neural Networks on Energy-Harvesting Edge Devices.**
P.-L. Sixdenier, M. Deutel, S. Wildermann, J. Teich. *7th Int. Workshop on IoT, Edge, and Mobile for Embedded Machine Learning (ITEM @ ECML-PKDD 2026)*.
- [3] **WiP Paper: Utility-Aware Transmission of Sensor Data on Energy-Harvesting IoT Gateways.**
P.-L. Sixdenier, J. Arockiaraj, S. Wildermann, J. Teich. *22nd Int. Conf. on Embedded Wireless Systems and Networks (EWSN 2025)*.
- [4] **GRES: Guaranteed Remaining Energy Scheduling of Energy-harvesting Sensors by Quality Adaptation.**
P.-L. Sixdenier, S. Wildermann, J. Teich. *13th Mediterranean Conference on Embedded Computing (MECO 2024)*.
- [5] **Hybrid Genetic Reinforcement Learning for Generating Run-Time Requirement Enforcers.**
J. Spieck, P.-L. Sixdenier, K. Esper, S. Wildermann, J. Teich. *21st ACM-IEEE Int. Symposium on Formal Methods and Models for System Design (MEMOCODE 2023)*.
- [6] **Seque: Lean and Energy-aware Data Management for IoT Gateways.**
P.-L. Sixdenier, S. Wildermann, M. Ottens, J. Teich. *IEEE Int. Conference on Edge Computing and Communications (EDGE 2023)*.
- [7] **RAVEN: Reinforcement Learning for Generating Verifiable Run-Time Requirement Enforcers for MPSoCs.**
K. Esper, J. Spieck, P.-L. Sixdenier, S. Wildermann, J. Teich. *4th Workshop on Next Generation Real-Time Embedded Systems (NG-RES 2023)*.
- [8] **SIDAM: A Design Space Exploration Framework for Multi-sensor Embedded Systems Powered by Energy Harvesting.**
P.-L. Sixdenier, S. Wildermann, D. Ziegler, J. Teich. *SAMOS XXII (Springer LNCS 2022)*.
- [9] **Towards an Autonomous, Power-Efficient Base Station for Sensor Data Collection.**
P.-L. Sixdenier. *2021 IEEE ACSOS-C*.

PROJECTS

- Personal:** Smart glasses (OLED HUD + BLE Android app), Pollution advisor (Arduino sensor crowdsourcing), Foot angle detector (ArUco computer vision).
- Academic:** Spiking Neural Network on FPGA (neuromorphic RTL modeling & simulation), “La futée” Smart Car (ADAS predictive driving assistance).

SKILLS & RECOGNITIONS

Technical: C++, C, VHDL, Python, Node.js, HTML/CSS • **DBMS:** MongoDB, Neo4J, PostgreSQL, MySQL
Embedded & FPGA: Arduino, STM32 (Keil μ Vision), Vivado, Quartus • **Tools:** Git, Docker, UML
Languages: French (Native), English (C1 / IELTS: 7 / TOEIC: 975), Spanish (B1)
Proud of: 2nd place at Renault Digital Hackathon 2019 • 2nd place at ENSEack 2020